



# Chapter 9

## A Blotting Paper Technique for the Screening of Chickpea Genotypes Against Dry Root Rot Disease

Athimoolam Durgadevi, Prachi Pandey, and Muthappa Senthil-Kumar 

### Abstract

Dry root rot (DRR) disease is a major threat to chickpea production across the world. This disease is caused by a soil-borne necrotrophic fungal pathogen, *Macrophomina phaseolina*. The use of disease-resistant varieties paves the way to conquer the disease spread. Though chickpea germplasm with rich genetic diversity is available around the world, its response to DRR is still unexplored. In turn, this demands screening and identification of resistant genotypes for crop protection against the disease. Here we describe an improved blotting paper technique for the large-scale screening of chickpea genotypes for DRR resistance. The method is quick, cost-effective, less labour-intensive, and thus optimized for high-throughput screening and can be efficiently used to screen a large number of chickpea genotypes for resistance against DRR.

**Key words** Chickpea, Dry root rot, *Macrophomina phaseolina*, Disease resistance, Screening, Blotting paper technique

---

## 1 Introduction

Chickpea (*Cicer arietinum* L.) is an economically important winter legume, usually cultivated in arid and semi-arid tropical regions under rain-fed conditions. It is a robust, dryland, self-pollinated crop, and has a deep root system with numerous and finer lateral roots along with a central taproot system. In agricultural fields, dry root rot (DRR) is a major threat to chickpea production [1]. A soil-borne necrotrophic fungus, *Macrophomina phaseolina* (formerly known as *Rhizoctonia bataticola*), causes DRR in chickpea. The typical DRR symptoms are chlorosis and drooping of leaves and petioles, leading to dead and dry plant. The infected plant leaves become straw yellow, and the taproot becomes dark, brittle, and

devoid of lateral roots [2, 3]. Microsclerotia can be seen in the roots and pith when vertically cut [4]. The DRR disease incidence is increased when the plants experience drought stress. The disease is more prevalent in rain-fed areas and is predominantly seen in the reproductive stage, that is, flowering to podding stages [2, 5].

Utilizing diverse genotypes or multilines with varying levels of resistance and susceptibility can significantly reduce the intensity and severity of disease outbreaks, offering better protection than single resistant varieties, which are more vulnerable to epidemics [6]. The use of cultivars that are resistant to DRR is the key to overcoming the disease. This is broadly the safest and an effective method for protecting the crops from the disease [7]. The large and diverse germplasm availability and knowledge of both plant and pathogen biology are important for successful screening. In molecular breeding programs, screening and identification of resistant and susceptible cultivars are essential for crop improvement. Easy, cost-effective, and high-throughput disease assays are crucial for the large-scale screening of the genotypes. Resistance screening against DRR is significant to protect the crop from the disease [8]. Till date, several methods like blotting paper technique, sick pot method, and sick plot screening have been developed and standardized to screen germplasm and develop disease-resistant cultivars [5, 9].

Among the different methods, blotting paper technique is simple and can be used to screen a large number of genotypes in a short time, unlike pot and field screening methods. This method can be executed simply with a liquid inoculum of the fungus and chickpea seedlings [9]. This chapter describes a high-throughput technique to screen a large number of chickpea genotypes for DRR resistance. This technique is quick, less tedious, and thus optimized for high-throughput screening. The seedlings can be inoculated with the fungus using dip inoculation or placing the agar plugs with the fungal sclerotia on the surface of the roots. Dip inoculation method is more reliable compared to agar plug method as it ensures the uniform attachment of the fungal mycelia to the roots.

---

## 2 Materials

- |                                        |                                                                                                                                         |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <b>2.1 Seed Material</b>               | Chickpea genotype seeds ( <i>see</i> <b>Note 1</b> ).                                                                                   |
| <b>2.2 Fungal Pathogen</b>             | <i>Macrophomina phaseolina</i> (ITCC 8635 with GenBank id, MH509971.1) ( <i>see</i> <b>Note 2</b> ).                                    |
| <b>2.3 Media and Chemical Reagents</b> | <ol style="list-style-type: none"> <li>1. Potato dextrose agar—PDA.</li> <li>2. Potato dextrose broth—PDB.</li> <li>3. Agar.</li> </ol> |

4. 2% Sodium hypochlorite for the surface sterilization of the chickpea seeds.
5. Soilrite, a combination of an equal ratio, that is, 1:1:1 of horticulture-grade expanded perlite, Irish peat moss, and exfoliated vermiculite, is used in the nursery to grow the seedlings.
6. Spirit.
7. CTAB (cetyl trimethyl ammonium bromide) extraction buffer: 3% CTAB, 100 mM Tris-HCl, pH 8.0, 2 M NaCl, 25 mM EDTA (ethylenediamine tetra acetic acid), 4%  $\beta$ -mercaptoethanol, 5% PVP (polyvinylpyrrolidone).
8. Chloroform.
9. Iso-amyl alcohol.
10. Isopropanol.
11. 70% ethanol.
12. TE buffer: 10 mM Tris-HCl, pH 8.0, 1 mM EDTA pH 8.0.
13. Agarose.
14. SYBR green.
15. 10% KOH solution
16. 1 M Tris-HCl, pH 7.0
17. 50 mM Tris-HCl, pH 7.0
18. WGA-FITC (wheat germ agglutinin-fluorescein isothiocyanate).

#### **2.4 General Materials and Equipment**

1. Petri dishes for the culturing of fungus.
2. Laminar airflow.
3. Falcon tubes (50 mL) for surface sterilization.
4. Vortex mixer.
5. Polythene bags with a dimension of 30 × 25 cm for raising the nursery.
6. 1000 mL conical flask for the preparation of broth culture
7. Autoclave for the sterilization of medium.
8. Sterilized glass or plastic funnel for filtering out fungal mycelia and microsclerotia.
9. Nylon mesh with a diameter of 1 mm for the filtration of fungal mycelia and microsclerotia.
10. Petri plates with a diameter of 20 cm.
11. Autoclaved RO (reverse osmosis) water.
12. Rectangular plastic trays with a dimension of 42 × 30 cm.

13. Autoclaved and oven-dried blotting papers.
14. Spray bottle.
15. Paintbrush.
16. Mortar and pestle.
17. Microcentrifuge tubes (2 mL and 1.5 mL).
18. Water bath/heating block.
19. Table-top centrifuge.
20. NanoDrop spectrophotometer.
21. Thermal cycler.
22. Autoclaved MQ water.
23. Nuclease-free water.
24. 96-well PCR plate
25. Fast RT-qPCR system.
26. Research stereomicroscope.
27. Fluorescence microscope.
28. Glass slides and coverslip.
29. Forceps.

---

### 3 Methods

#### **3.1 Preparation of Fungal Plate**

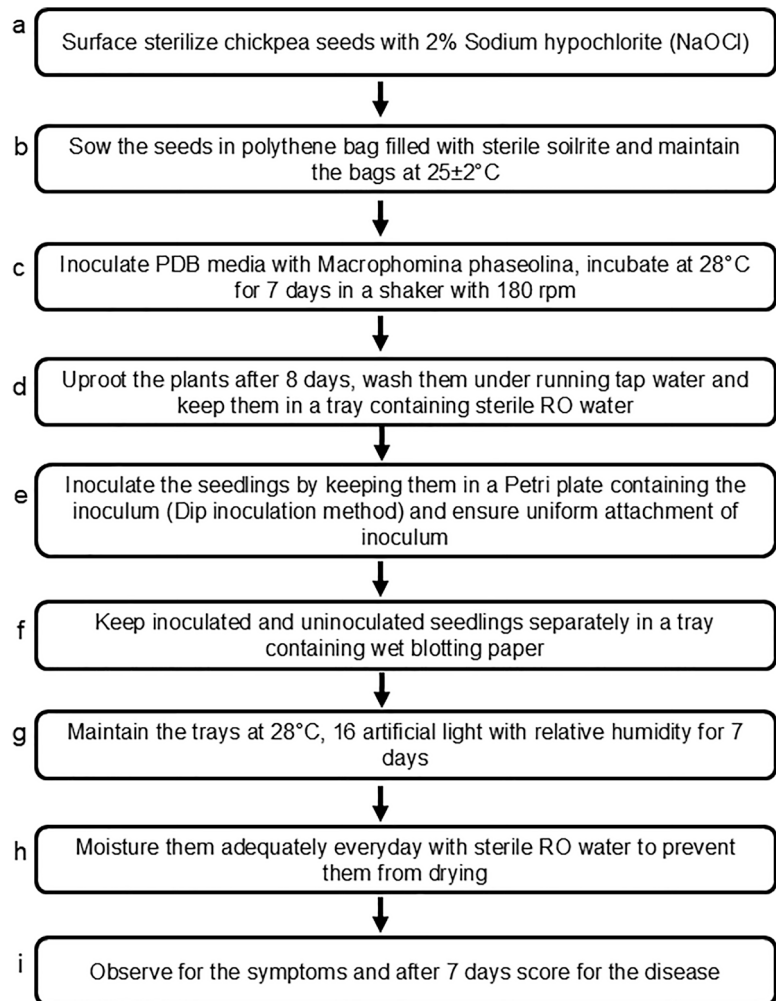
1. Prepare 100 mL of PDA medium in a conical flask and autoclave it at 120 lbs. for 15 min.
2. Let the medium cool down, pour it into the plates and let the medium solidify.
3. Cut the tip of hyphal growth and inoculate it in the fresh PDA plates.
4. Incubate the plates at 28 °C for 5 days.

#### **3.2 Raising of Seedlings**

Select chickpea genotype seeds with uniform shape and size (Fig. 2a).

##### **3.2.1 Surface Sterilization of Seeds**

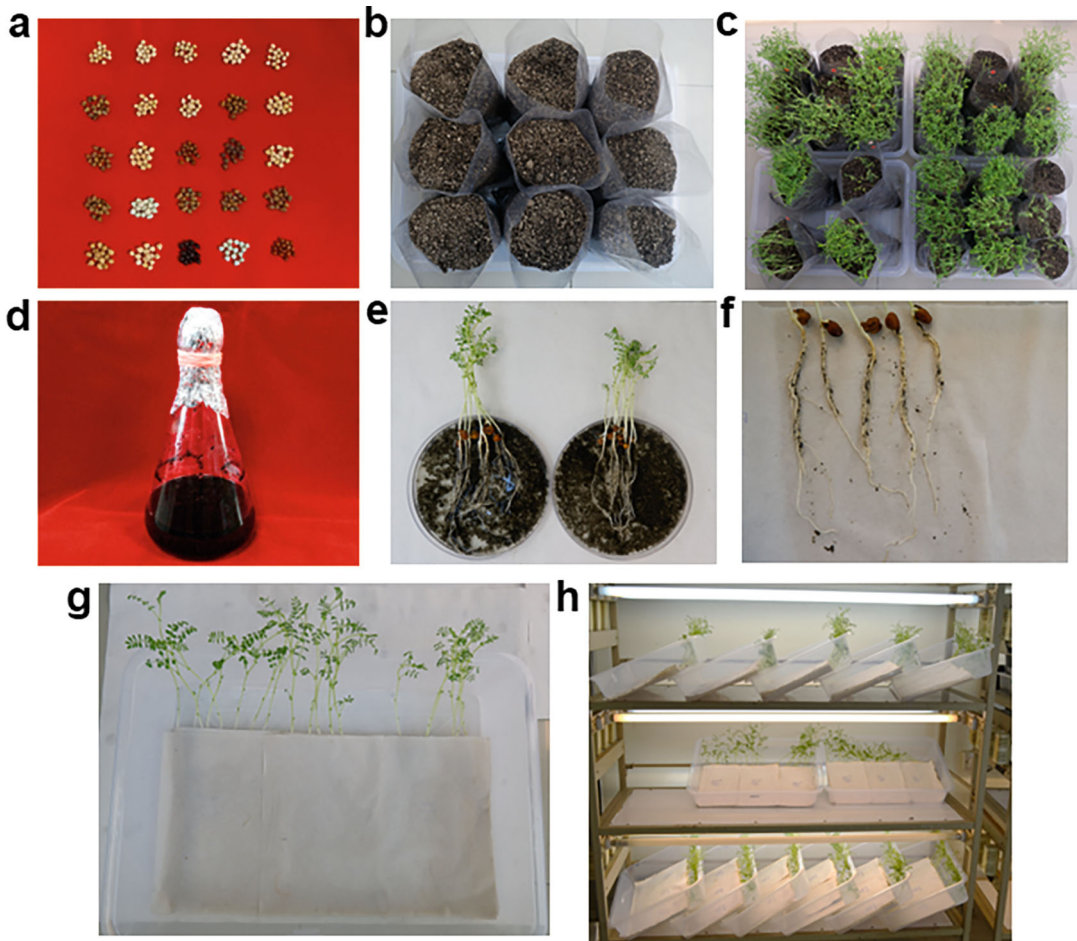
1. Take the seeds in Falcon tubes and wash them with autoclaved RO water for a minute, three time, to remove the dust particles under laminar airflow.
2. Treat the washed seeds with 2% sodium hypochlorite (NaOCl) under intermittent shaking using a vortex mixer for about 2 min (Fig. 1a).
3. Then rinse the seeds with autoclaved RO water to remove the traces of NaOCl.



**Fig. 1** Flowchart depicting the steps involved in high-throughput screening of chickpea genotypes for DRR resistance using blotting paper technique

### 3.2.2 Sowing

1. Fill polythene bags with autoclaved soilrite (Fig. 2b).
2. Sow the surface sterilized seeds in the bags with soilrite and keep them at  $25 \pm 2^\circ\text{C}$ , 16 h photoperiod with 70% humidity (Fig. 1b).
3. Uproot the plants 7 days after sowing and gently wash the roots (Fig. 1d) under running tap water to remove the soil particles completely without damaging the roots (Fig. 2c). Then keep the plants in a tray containing autoclaved RO water to allow infection (*see Note 3*).



**Fig. 2** Detailed methodology of blotting paper technique for DRR disease screening in chickpea. The seeds were collected (a) and surface-sterilized using sodium hypochlorite and rinsed with sterile RO water. The surface-sterilized seeds were sown in polythene bags filled with soilrite (b) and maintained at 28 °C, 16 h photoperiod with an RH of 60–70%. After 8 days (c), the plants were uprooted and washed under tap water gently to remove the soil particles. PDB media was prepared and inoculated with actively growing *Macrophomina phaseolina* and incubated at 28 °C at 180 rpm in dark. After 7 days (d), the fungal mycelia and microsclerotia were filtered out and dissolved in autoclaved RO water. Then the plant roots were dipped (e) in the inoculum for a min with intermittent shaking. The inoculated plants were then kept in a tray containing wet blotting sheet (f). The blotting sheet was then folded into half to cover the roots (g). Inoculated and uninoculated seedlings were kept in separate trays. Then all the trays were maintained at 28 °C, 16 h photoperiod with a relative humidity of 60–70% for 7 days (h). The plants were adequately moisturized throughout the 7 days and observed for disease symptoms. After 7 days, the plants were scored for the disease

### 3.3 Preparation of Liquid Inoculum

1. Prepare 300 mL of PDB medium in four 1000 mL conical flasks, and autoclave them at 120 lbs. for 15 min.
2. Let the medium cool down completely, then inoculate the media with 9 mm discs of agar plugs from a 5-day-old fungal agar plate with actively growing *M. phaseolina* culture (Fig. 1c).

3. Incubate the conical flasks at 28 °C in a shaker at 180 rpm in the dark for 7 days.
4. After 7 days (Fig. 2d), using an autoclaved glass or plastic funnel with multiple layers of nylon mesh with a 1 mm diameter pore size to filter out the fungal mycelia and microsclerotia.
5. Wash the mycelia thoroughly with autoclaved RO water to remove the medium completely and then use it as an inoculum.

### 3.4 Preparation of Blotting Sheets

1. Cut the blotting papers into pieces with a dimension of 30 × 24 cm in sufficient numbers to meet all the replications in large-scale screening.
2. Pack the sheets in a bunch in autoclavable polythene bags, and autoclave at 121 lbs. for 15 min.
3. Keep the blotting sheets in a hot air oven for drying till the moisture gets completely removed (*see Note 4*).

### 3.5 Inoculation

#### 3.5.1 Dip Inoculation

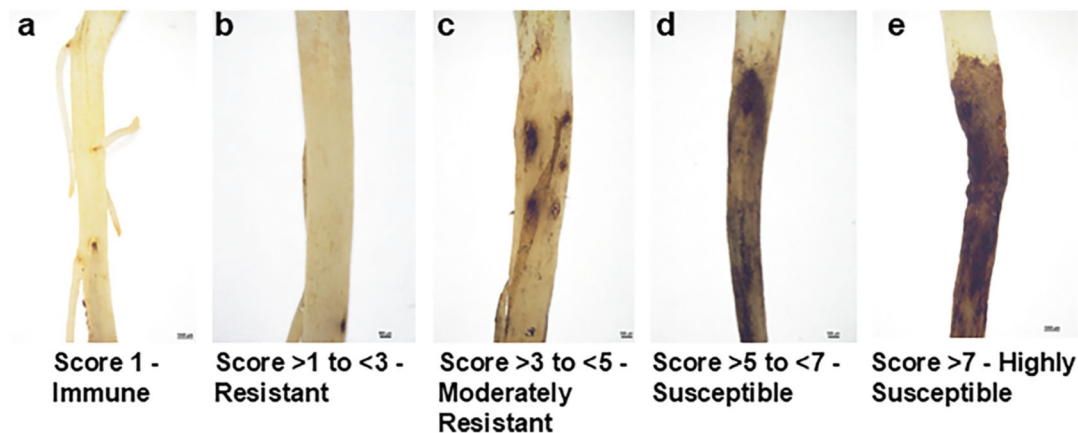
1. Dissolve the washed mycelia (**step 5**, Subheading 3.3) in autoclaved RO water (400 mg/mL) in multiple petri plates with a diameter of 20 cm (*see Note 5*).
2. Dip the uprooted plants (**step 3**, Subheading 3.2.2) in the inoculum (Fig. 1e) for a minute in continuous movement to ensure uniform attachment of fungal mycelia on the roots (Fig. 2e).

#### 3.5.2 Agar Plug

1. Take a 5-day-old *M. phaseolina* plate.
2. Cut 2 cm blocks using pipette tips and keep them on the surface of the roots from the point of seed attachment after transferring the seedlings to blotting paper (Subheading 3.5).

### 3.6 Transfer of Seedlings to Blotting Paper

1. Take a clean tray, wipe it thoroughly with spirit, and pour sterile (autoclaved) RO water, then keep the autoclaved blotting sheets (**step 3**, Subheading 3.4) in the water (*see Note 6*).
2. Arrange the wet blotting sheets in a clean, sterilized tray in two layers.
3. Place the inoculated seedlings side by side (five seedlings/replicate) on a wet blotting paper (Fig. 2f), and fold the sheet into half so that only the roots will be covered and the green foliage will be left outside (Fig. 2g).
4. Place the folded blotting papers with the seedlings of the same genotype (three replicates/genotype) one over the other.
5. Keep the plants of three different genotypes with all the replicates in a single tray.



**Fig. 3** Representative root images for disease scoring based on the severity of dry root rot. Root images (2 cm from the point of seed attachment) taken 7 days after infection using stereomicroscope SMZ25. Image shows (a) no infection on roots (score 1/immune), (b) very few lesions on roots (score > 1 to <3/resistant), (c) roots on the lesions are clear but small (score > 3 to <5/moderately resistant), (d) lesions on roots are clear and visible (score > 5 to <7/susceptible), (e) roots are infected and completely discolored (score > 7/highly susceptible). Scale bar—1000  $\mu\text{m}$

6. Place control and inoculated seedlings in separate trays (Fig. 1f), and maintain them at  $28 \pm 2^\circ\text{C}$ , 16 h photoperiod (Fig. 1g) with a relative humidity of 70% for 7 days (Fig. 2h).
7. Water all the trays adequately to prevent the roots from drying (Fig. 1h) (see Note 7).
8. Observe the plants after 7 days for DRR disease symptoms and score for the disease.
9. Keep a DRR susceptible and a resistant check with every batch (see Note 8).

### 3.7 Scoring for the Disease

1. Observe the plants for leaf yellowing and root necrotic symptoms (Fig. 1i) and, after 7 days, score for the disease (Fig. 3).
2. Represent the disease severity on a scale of 1–9, and give a qualitative classification based on the rating scale (Table 1).
3. At the end of the infection period, gently remove the fungal mycelia using autoclaved RO water and a paintbrush from each root surface to see the infection visibly. Assess the degree of disease severity using the disease rating scale given (Fig. 4).

### 3.8 In Planta Fungal DNA Quantification

#### 3.8.1 DNA Isolation Using CTAB Method

1. Take 250 mg of frozen root sample and add 2 mL of CTAB extraction buffer.
2. Incubate at  $70^\circ\text{C}$  for 1 h, and cool at room temperature.
3. Prepare chloroform: Isoamyl alcohol in a ratio of 24:1, and add equal volume of this to all the tubes.

**Table 1**  
**Disease severity on 1–9 rating scale for dry root rot of chickpea**

| Rating/score | Observation/symptoms                                            | Disease reaction     |
|--------------|-----------------------------------------------------------------|----------------------|
| 1            | No infection on roots                                           | Immune               |
| >1 and < 3   | Very few small lesions on roots                                 | Resistant            |
| >3 and < 5   | Lesions on roots clear but small; new roots free from infection | Moderately resistant |
| >5 and < 7   | Lesions on roots; many new roots generally free from lesions    | Susceptible          |
| >7           | Roots infected and completely discolored                        | Highly susceptible   |

4. Mix the tubes well and centrifuge at 10,000 g for 10 min at room temperature.
5. Transfer the aqueous phase to a fresh tube without disturbing, add 0.6 volume of isopropanol.
6. Centrifuge the tubes at 10,000 g for 10 min at 4 °C.
7. Discard the supernatant, wash the pellet with 70% ethanol, and centrifuge at 10,000 g for 10 min at 4 °C.
8. Air-dry the pellet and dissolve it in 30 µL TE buffer.
9. Measure the quality and quantity of genomic DNA using 0.8% agarose gel and NanoDrop spectrophotometer.
10. Confirm the presence of *M. phaseolina* using PCR with specific primers (*see Note 9*).

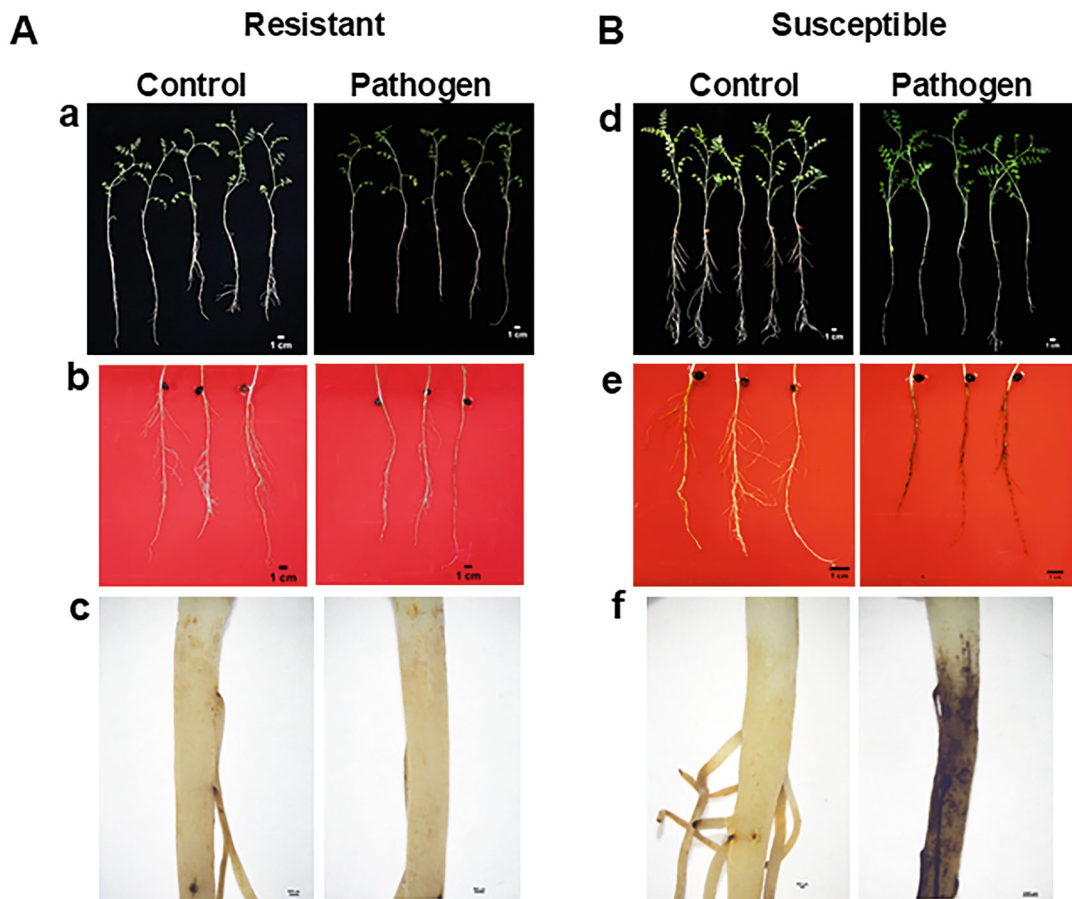
### **3.9 Quantification of *M. phaseolina* Using qPCR**

#### **3.9.1 DNA Concentration Normalization**

1. Normalize the DNA concentration to 50 ng/µL.

#### **3.9.2 Generation of Standard Calibration Curve**

1. Collect the fungal mycelia from a 5-day-old fungal plate and isolate fungal genomic DNA using the protocol mentioned above.
2. Dilute the pure fungal genomic DNA to tenfold serially (from 100 to 1 pg/µL) with sterile autoclaved MQ.
3. Prepare the master mix of 9 µL containing 5 µL KAPA SYBR FAST Master Mix (2X) ABI Prism™ and 1 µL of each of the primers (Mph-F and Mph-R) (450 nm) and 3 µL nuclease-free water.
4. Aliquot 9 µL master mix to all the wells of the 96-well PCR plate.



**Fig. 4** DRR disease symptoms in resistant and susceptible chickpea genotypes under blotting paper technique. The plants were scored after 7 days of infection. Images shows (a) phenotype of resistant genotype, (a) whole plant image of the both control and pathogen, (b) root images of control and pathogen-infected plants, (c) single root image taken 2 cm from the point of seed attachment under research stereomicroscope SMZ25 of both control and pathogen-infected plants of resistant genotype. (b) phenotype of susceptible genotype. (d) whole plant image of both control and pathogen, (e) root images of control and pathogen-infected plants, (f) single root image taken 2 cm from the point of seed attachment under research stereomicroscope SMZ25 of both control and pathogen-infected plants of susceptible genotype. The pictures were captured using Nikon D5100. Scale bar = 1 cm, 1000  $\mu$ m

5. Add 1  $\mu$ L of genomic DNA of *M. phaseolina* to the plate, from 1 pg dilution to 100 ng dilution for standard curve preparation. Include three replicates per dilution.
6. Add 1  $\mu$ L of DNA isolated from the root sample (50 ng/ $\mu$ L) to each well separately.
7. Add 1  $\mu$ L water to three wells for the no template control (NTC).
8. Load the 96-well PCR plate containing samples into the 7900HT Fast RT-qPCR system.

9. Adjust the program to include an initial denaturation step for 5 min at 95 °C, followed by 40 cycles of denaturation at 95 °C for 10 s and annealing at 65 °C for 35 s, followed by heating to 95 °C, cooling to 65 °C for 1 min, and then heating to 97 °C at a rate of 1 °C/5 s [10].
10. Report the results as the mean C q value. Additionally, the target can be reported as fungal concentration per defined unit of the template.
11. Determine the concentration of fungal DNA present from the linear equation of the amplification using the target fungal gene C q values.
12. Generate the linear equation from the standard curve of known starting DNA concentrations containing the target gene (*see Note 10*).

### 3.10 Microscopy

#### 3.10.1 Staining

1. Visualize the roots under a research stereomicroscope to observe the necrosis.
1. Cut the roots into 2 cm from the point of seed attachment.
2. Place them in a microcentrifuge tube (2 mL) containing 10% KOH for root clearing and keep them on a shaker for 48 h at room temperature (*see Note 11*).
3. Remove KOH and neutralize the root samples with 1.5 mL of 1 M Tris-HCl (pH 7.0).
4. Replace the buffer with 50 mM Tris-HCl (pH 7.0).
5. Place the root sample in a clean glass slide and add 100 µL of WGA-FITC (50 µg/mL) (*see Note 12*) over the root.
6. Place a coverslip on the root and press gently with forceps.
7. Remove excess stain using tissue paper and observe the roots under a fluorescence/confocal microscope with a bandpass of 485/535 nm [11].

---

## 4 Notes

1. The seeds can be procured from ICRISAT (International Crops Research Institute for the Semi-Arid Tropics), Patancheru, Hyderabad, Telangana.
2. The soil-borne necrotrophic fungus was isolated from the roots of an infected plant collected from a field in NIPGR (28.5307 °N, 77.1665 °E), New Delhi. The pure culture was obtained using the hyphal tip method [7]. The same can be procured from ITCC.

3. After uprooting, do not keep the plants in water for more than 30 min. Once uprooted, keep them in the blotting sheet within 3 h.
4. Blotting sheets should be completely dried; otherwise, it may lead to secondary infection.
5. Mix 20 g of filtered fresh mycelia with 50 mL of autoclaved RO water to prepare 400 mg/mL (40%) inoculum.
6. To avoid secondary infection, use a freshly opened autoclaved polythene bag with blotting sheets for every batch.
7. Watering more can cause the rotting of the roots, and less watering may lead to the drying of roots. Spray water frequently (three times/day) to avoid these problems.
8. CSJ 515 (DRR-resistant) and BG 212/L550 (DRR-susceptible) can be used as check varieties.  
Mph-F—AAACACACTTTTCGCACTCCTGC.  
Mph-R—TATGCTCGCAGAGAAGAACACGA.  
are specific primers for the confirmation of *M. phaseolina*.
9. Ensure no amplification with target primers in the control plant samples or the NTC that indicate the absence of *M. phaseolina* DNA in the uninfected roots and no cross-contamination or primer dimers.
10. Root clearing helps in removing the cytoplasmic contents of the cell, which in turn ensures better visualization of the fungal mycelia.
11. Prepare 50 µg/mL working solution from a 1 mg/mL stock solution by diluting it.

---

## Acknowledgments

Projects at the M.S.K lab are supported by the National Institute of Plant Genome Research core funding and DBT multi-institutional project (BT/Ag/Network/Chickpea/2019-20) “**Characterization of chickpea germplasm resource to accelerate genomics-assisted crop improvement**,” under mission program of “Characterization of genetic resources.” We thank Mr. Rahim H Tarafdar and Mr. Sunder Solanki for their help in the laboratory. We acknowledge DBT-eLibrary Consortium (DeLCON) and NIPGR Library for providing access to e-resources and NIPGR Plant Growth Facility for plant growth support/space.

## References

1. Sharma M, Ghosh R, Pande S (2015) Dry root rot (*Rhizoctonia bataticola* (Taub.) Butler): an emerging disease of chickpea – where do we stand? *Arch Phytopathol Plant Protect* 48:13–16
2. Sharma M, Ghosh R, Sharma TR, Pande S (2012) Intra population diversity in *Rhizoctonia bataticola* causing dry root rot of chickpea (*Cicer arietinum* L.) in India. *Afr J Microbiol Res* 6. <https://doi.org/10.5897/AJMR12.954>
3. Rai A, Irulappan V, Senthil-Kumar M (2022) Dry root rot of chickpea: a disease favored by drought. *Plant Dis* 106:346–356. <https://doi.org/10.1094/PDIS-07-21-1410-FE>
4. Sinha R, Irulappan V, Mohan-Raju B et al (2019) Impact of drought stress on simultaneously occurring pathogen infection in field-grown chickpea. *Sci Rep* 9:5577
5. Nene YL, Haware MP, Reddy MV (1981) Chickpea diseases: resistance-screening techniques. Information Bulletin No. 10. Patancheru, A.P., India: International Crops Research Institute for the Semi-Arid Tropics
6. Zhu Y, Chen H, Fan J et al (2000) Genetic diversity and disease control in rice. *Nature* 406:718–722
7. Johnson R (1992) Past, present, and future opportunities in breeding for disease resistance, with examples from wheat. In: Johnson R, Jellis GJ (eds) *Breeding for disease resistance. Developments in plant pathology*, vol 1. Springer, Dordrecht
8. Mirchandani R, Irulappan V, Chilakala AR, Senthil-Kumar M (2023) Dry root rot disease: current status and future implications for chickpea production. *Proceedings of the National Academy of Sciences, India Section B*
9. Irulappan V, Senthil-Kumar M (2021) Dry root rot disease assays in chickpea: a detailed methodology. *J Vis Exp* 167:e61702
10. Irulappan V, Mali KV, Patil BS, Manjunatha H, Muhammad S, Senthil-Kumar M (2021) A sick plot-based protocol for dry root rot disease assessment in field-grown chickpea plants. *Appl Plant Sci* 9:e11445
11. Irulappan V et al (2022) Drought stress exacerbates fungal colonization and endodermal invasion and dampens defense responses to increase dry root rot in chickpea. *Mol Plant-Microbe Interact* 35(7):583–591